Artificial Intelligence and Its Carbon Footprint

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***Abstract*— Artificial Intelligence (AI) is transforming industries, but its rapid expansion comes with environmental costs. AI systems consume vast amounts of energy and contribute significantly to global carbon emissions. Training a single large model, such as GPT-3, emits approximately 284 metric tons of CO2 , equivalent to the lifetime emissions of five cars. AI- driven data centers currently account for 2.5%–3.7% of global greenhouse gas emissions. As AI adoption grows, balancing its technological benefits with environmental responsibility is urgent.**

This paper synthesizes research on AI's carbon footprint and explores strategies to mitigate its impact while fostering inno- vation. Several solutions can drive sustainable AI development. Optimizing AI models through techniques like pruning, quantiza- tion, and knowledge distillation can reduce computational costs by 40%–50%. Transitioning data centers to renewable energy sources could lower emissions by 30%–40%. Policy interventions, such as AI carbon taxes and industry sustainability standards, can encourage companies to adopt greener practices. Companies like Google and Microsoft are already leveraging AI for energy- efficient operations, demonstrating the feasibility of sustainable AI growth.

A sustainable AI future requires collaboration among stake- holders. Industry leaders, policymakers, and researchers must work together to integrate green computing principles into AI development. By investing in energy-efficient algorithms, responsible hardware design, and cleaner energy sources, we can reduce AI's environmental impact. This research underscores the urgency of immediate action. AI must evolve responsibly, ensuring technological progress does not come at the cost of environmental degradation.

KEYWORDS: Artificial Intelligence, Carbon Footprint, Sustainability, Green Computing, Energy Efficiency, Introduction

I. INTRODUCTION

Artificial Intelligence, or AI, is one of the most rapidly developing technological advancements today. Its seemingly overnight expansion is comprehensive as it provides many convenient, previously unseen features. Unfortunately, this innovative technology comes with a significant ecological con- sequence. Large-scale AI models, such as ChatGPT and Google Gemini, consume substantial energy. This phenomenon is attributed to various factors, which will be discussed in greater detail. The growing concern over the concealed carbon footprint associated with AI systems necessitates evaluation. Research estimates that training a single large AI model can result in over 280 metric tons of carbon dioxide emissions [1]. Artificial intelligence is often labeled as "clean", but its environmental consequences cannot be ignored. Despite the continued rapid public growth of AI, there is also an increasing media and academic attention on the carbon footprint generated by AI systems. Therefore, it is crucial to understand the factors contributing to the substantial emissions produced by AI. The utilization of artificial intelligence necessitates substantial energy consumption for various components, including data farms and the training and deployment of its systems. Data centers, primarily powered by fossil fuels, are among the primary systems utilized for generating and processing artificial intelligence. [3] Furthermore, there is significant regional inequality in terms of carbon burden, primarily due to companies locating data centers in more rural areas.

To address the environmental impact of AI, several initiatives are being undertaken. Green AI techniques, such as pruning, quantization, and efficient model design, are being employed to reduce energy consumption. Additionally, there is an increased adoption of renewable energy sources, cloud computing, and restructuring. Carbon tracking tools, like Eco2AI, are being used to promote transparent reporting [4]. Furthermore, businesses are adopting corporate

sustainability pledges to publicly identify their efforts and initiatives.

Figure 1. Construction near Princess Elizabeth Island (World's first AI island, 28 miles long energy center) [2].

A. Stakeholders

Several major stakeholders are involved in defining the topic of artificial intelligence's impact on the environment. These stakeholders include technology companies such as OpenAI, Google, and Microsoft. It also includes governments and regulatory agencies that acknowledge the use of artificial intelligence [5] and its effects on the general public. Finally, it includes client scientists and environmentalists; researchers and developers; and the general public and future generations.

The impact of artificial intelligence on the environment is significant for several reasons. Firstly, it is a rapidly growing technology, as previously mentioned, and therefore its environmental impact is likely to continue growing in the coming years. Second, unsustainable artificial intelligence expansion generates a significant potential threat to ongoing global climate goals, as climate change is a major topic of discussion today [6]. Last, there are a number of ethical concerns regarding the creators and developers of artificial intelligence and their current strategies identified for ensuring long-term environmental safety [7].

Given these considerations, two major questions remain: should artificial intelligence performance be prioritized over environmental causes? Who is responsible for reducing artificial intelligence's impact, carbon footprint, and other environ- mental effects: the creators, users, or regulators? This paper seeks to explore these questions further.

II. Evaluation Criteria

To have a strong judgment on AI's full environmental im- pact, there must be carbon reduction strategies set in place us- ing clear and consistent criteria. The environmental evaluation of AI systems can be guided by six core questions designed to assess the overall significance of carbon reduction strategies in this field [8]. These questions will also help determine whether AI climate initiatives are genuinely contributing to a common climate goal, or just simply appearing to do so.

A. Six Key Questions

The term,'relevance,' confirms that programs address the main causes of AI-related emissions, such as the energy that is required to train large-scale machine learning models and power vast data centers. Addressing these vast contributions is substantially more successful than addressing smaller ones, therefore determining relevance is an important first step in figuring out any solution. The effectiveness of a project is determined by whether it saves measurable quantities of energy, or lowers greenhouse gas emissions. Approving more efficient methods or converting data centers to renewable energy must be supported by verified evidence.

The third evaluation criteria, impact, looks to more than the immediate results to arbitrate long-term environmental effects. A plan may reduce emissions today, but its true impact is seen when it directs AI change towards long-term and climbable sustainability.

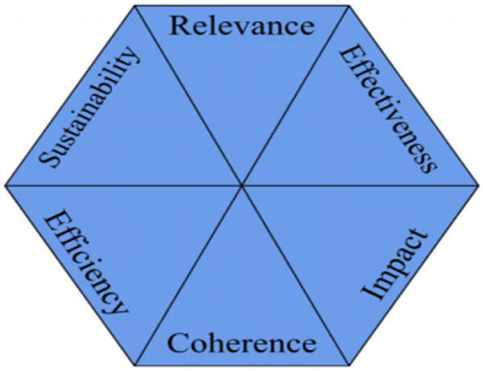

Figure 2. Six Key Questions for carbon reduction.

Even the greatest attempts might fail if they lack a wide perspective on the topic. Coherence estimates how well a program fits into the broad text of global climate goals and ethical development. Carbon reduction should not come at the price of ethical principles. These principles may consist of justice, openness, and accessibility. Aligning environmental policies with these overall goals would strengthen the validity and worth of the ethical situation at hand.

Efficiency will refer to how effectively resources are used to meet sustainability goals. Optimizing productivity while minimizing waste is critical in a location where processing power, time, and money are limited. Efficient strategies, like updating current models and eliminating duplicate training, would overall help reduce emissions while maintaining solid performance. Lastly, sustainability gauges whether an action's advantages will carry over as AI systems expand in size and complexity. Some strategies may work for existing models, however most would fail when

escalated. Effective assessment should foresee future issues and encourage approaches that would be sustainable throughout time.

These six criteria; relevance, effectiveness, impact, coherence, efficiency, and sustainability; make up an all-inclusive planning for determining AI technology's environmental du- ties. Researchers and developers would confirm that as long as the efforts to decrease AI's carbon footprint are carried out correctly, then they are extremely relevant to the overall goal.

III. Methodology

A This study employs a dual-method approach, combining a systematic literature review with quantitative data synthesis. The overarching objective is twofold: (1) to quantify the energy consumption associated with modern AI models, and (2) to identify and assess the effectiveness of methods aimed at reducing such energy usage.

A. Research Questions

The investigation is guided by the following research questions:

- RQ1: What is the magnitude of energy consumption in recent large-scale AI models? - RQ2: What strategies have been proposed to improve energy efficiency in AI, and how effective are these strategies?

These questions reflect the core aim of the study—to understand not only the scale of energy use in AI but also the potential of technical interventions to mitigate its environ- mental footprint.

B. Literature Selection Process

To address the research questions, we implemented a structured and transparent selection process for identifying relevant academic literature. Six peer-reviewed papers were ultimately selected, representing a mix of empirical investigations and systematic reviews that collectively define the current land- scape of research on AI energy usage and Green AI practices.

The selection was guided by the following criteria:

- Inclusion criteria: Studies that either provide direct empirical data on the energy consumption of AI systems or propose and benchmark methods for reducing such consumption. - Exclusion criteria: Studies whose sole focus is the use of AI for sustainability applications (e.g., climate modeling), without evaluating the energy demands of the AI systems themselves.

The selected works included detailed energy and carbon assessments of specific AI models, such as GPT-3, as well as broader literature reviews on Green AI techniques. Publications were sourced from leading digital libraries and conferences, including arXiv, IEEE, NeurIPS, CVPR, WIREs, and ICML, ensuring both credibility and relevance.

C. Data Extraction and Analysis

For each selected study, we systematically extracted key variables that facilitate comparative evaluation. These variables included:

- The type of AI model analyzed (e.g., Transformer, Convolutional Neural Network) - Measured energy consumption (e.g., in kilowatt-hours [kWh] or megawatt-hours [MWh])

- Reported carbon emissions (e.g., in metric tons of CO2 equivalent, tCO2 e, where available) - Optimization techniques applied (e.g., sparsity, hardware acceleration, algorithm-level tuning) - Documented energy savings (reported in absolute values or as a percentage reduction) To enable coherent com- parison across studies, the extracted data were organized into standardized tables and classified into four thematic domains:

1. Hardware and Infrastructure: Including the use of specialized chips (e.g., TPUs) and energy-efficient data center configurations 2. Model Architecture: Emphasizing structural innovations such as sparse activation and neural architecture search

3. Software and Algorithmic Optimization: Covering techniques such as hyperparameter tuning, model compression, and reduced-precision computation 4. Deployment Strategies: Including edge computing, quantization, and inference-time optimizations This categorization captures the critical points of intervention along the AI development and deployment pipeline where energy savings can be realized.

D. Evaluation of Effectiveness

The effectiveness of each strategy was evaluated based on the reported energy savings in the primary sources. Where necessary, data were normalized—typically by converting results into percentage reductions relative to baseline configurations—to facilitate cross-study comparison.

Although methodological variations exist among the selected studies, a consistent pattern emerges: many of the proposed strategies lead to substantial energy reductions, with several exceeding a 50% decrease. This trend underscores the importance and feasibility of incorporating energy-aware design choices into the development life cycle of AI systems.

IV. Results

AI has led to many significant technological advancements, however, the exponential growth of large-scale models contributes to an increasing environmental burden. As these technologies become more advanced, so do their environmental impacts. Particularly, when identifying energy consumption and its carbon emissions.

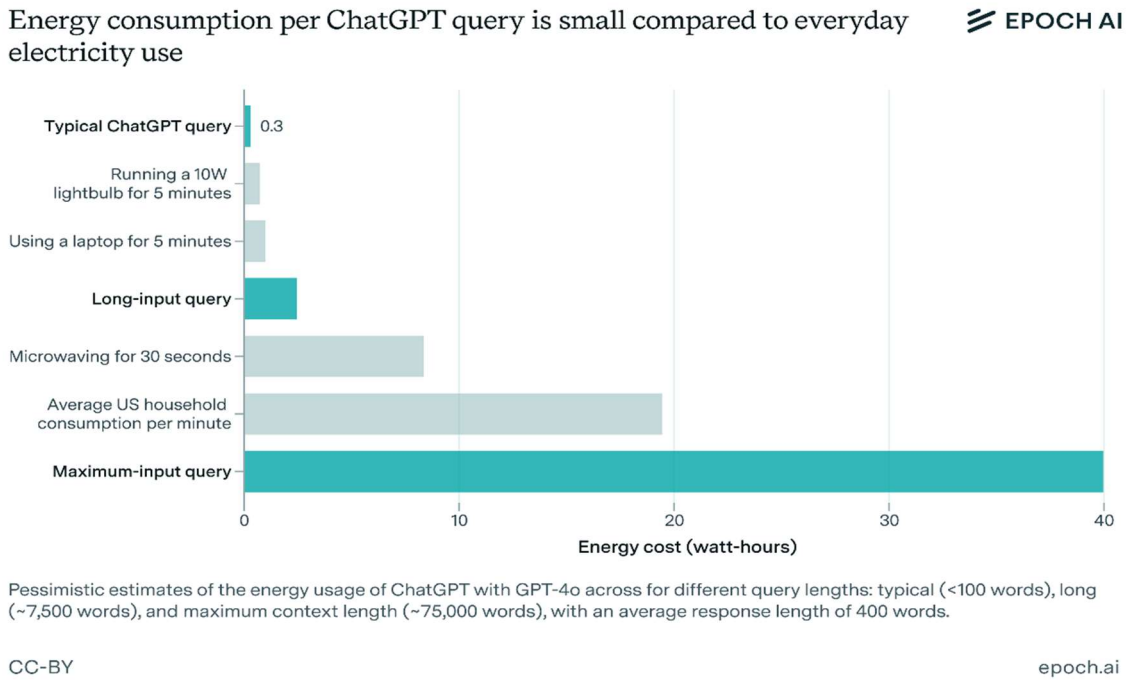

Figure 3. Energy consumption due to Chat GPT [12].

To begin, AI requires large amounts of energy to train models. This includes programs such as GPT-3, Google BERT, etc. that rely on heavy amounts of fossil fuels. It is estimated that OpenAI's GPT-3 uses approximately 1.287 gigawatt–hours of electricity, which is equivalent to the annual energy consumption of around 120 homes in the United States [9]. Another example is Google's BERT model, which emits approximately 1400 pounds of carbon dioxide. [10]

In addition to this evidence, ecological risks, regional bias are also prevalent when identifying AI's geological impact on the environment. This is prevalent in areas, such as Virginia, where there are large amounts of data centers, utilizing large amounts of renewable resources, such as coal and gas [11]. This additionally contributes to the overwhelming amount of carbon dioxide being produced. Around the world, data centers use about one percent of the world's electricity in terms of artificial intelligence [11]. To illustrate the gravity of this notion, a single AI model has the potential to utilize dozens of training and fine-tuning trials. All in all, this data provides a clear understanding of the risks Artificial Intelligence poses to the climate and additionally recognizes the need for effective strategies to combat this epidemic.

V. Conclusion

As AI evolves at the pace of two clicks, its implications for the environment are becoming difficult to mask. The explosion in AI models — particularly in natural language processing (NLP) and image generation — has not come free of an even steeper rise in energy consumption and carbon emissions alongside it. And this trend poses an urgent problem: AI has real social and technological value, but its growing carbon footprint has the potential to muddy those dividends unless we act upon it.

Without a social structure around how data can be managed, this has approached is on the cusp of exceeding the societal good this technology with an environmental impact. These costs account for the energy used in training and deploying models, as well as the aggregate usage of infrastructure (both data centers and hardware manufacturers and constant updates). Given the pervasive integration of AI systems into daily life and business practice, long-term viability is as urgent a priority as high performance.

So, the balance between performance and sustainability is going to be critical. That involves optimizing models to run less energy-hungry systems, moving to more renewable- powered infrastructure, and increasing the transparency of the processes purposes through emissions tracking tools caused and civil society collaboration between government, researchers and industry. There is enormous potential for AI to

power climate solutions — if its own development is managed responsibly. Once more, for AI to become a socially responsible tool, it needs to change: its priorities should be recontextualized from power-maxed at any cost to purposive-optimal design that takes environmental stewardship into account.

ACKNOWLEDGMENT

We would like to formally express our appreciation to those who contributed to the growing conversation around the environmental impact of artificial intelligence. A collective effort is needed to call attention to the urgent need for sustainability within the tech industry. As AI continues to advance, so must our commitment to ethical innovation. One that respects not just human progress, but the planet we depend on.

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